Liam Crotty
ljc233

Problem 32 Solution

32) Rolling cylinder A uniform cylinder with mass m and radius r rolls without slip outside and on top of a fixed cylinder with radius R. Gravity g pulls down.

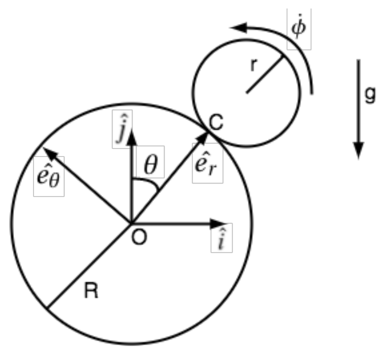

Figure 1: Diagram of the system with the reference frames and angles being measured.

(a) Are the full non-linear differential equations the same as those of the inverted pendulum? If so, or not, explain why this is expected.

The equation of motion for a pendulum is $\ddot{\theta} = -\frac{g}{l} sin(\theta)$.

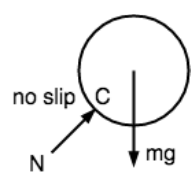

Figure 2: Free body diagram of the top cylinder.

AMB@C:

$$\Sigma \bar{M}_{/C} = \dot{\bar{H}}_{/C} \tag{1}$$

$$\bar{r}_{G/C} \times -mg\hat{j} = \bar{r}_{G/C} \times m\bar{a}_G + I\ddot{\phi}\hat{k} \tag{2}$$

The radius of G to C is just $\bar{r}_{G/C} = r\hat{e}_r$.

$$r\hat{e}_r \times -mg\hat{j} = r\hat{e}_r \times m\bar{a}_G + I\ddot{\phi}\hat{k} \tag{3}$$

The above equation depends on a moment of inertia term where as the pendulum equation does not. This means the two equations will not be the same. This is expected because the pendulum equation treats the mass as a point mass and not a rigid body.

(b) Assume the rolling cylinder is released from rest almost exactly at the top: $\theta = 0^+$ At what angle will it lose contact. Find an analytic formula for this. [Hint: use energy conservation.] Extra: if you like, compare the result with a numerical solution of the ODEs using 'events'.

Start by using conservation of energy and getting an expression for the potential and kinetic energy.

$$PE = mgh = mg(R+r)cos(\theta) \;\; assuming\, 0\, at\, O \tag{4}$$

$$KE = \frac{1}{2}mv_G^2 + \frac{1}{2}I\dot{\phi}^2 \tag{5}$$

To find the kinetic energy we find the kinematics of the system.

$$\bar{r}_G = (R+r)\hat{e}_r \tag{6}$$

$$\bar{v}_G = (R+r)\dot{\theta}\hat{e}_\theta \tag{7}$$

$$\bar{a}_G = (R+r)\ddot{\theta}\hat{e}_\theta - (R+r)\dot{\theta}^2\hat{e}_r \tag{8}$$

using the kinematics the kinetic energy can be solved for in relation to θ.

$$KE = \frac{1}{2}m(R+r)^2\dot{\theta}^2 + \frac{1}{2}I\dot{\phi}^2 \tag{9}$$

Summing the kinetic energy and potential energy we can get a relation for the total energy at any angle $\dot{\theta}$ and angular velocity $\dot{\phi}$.

$$E^{TOT} = mg(R+r)cos(\theta) + \frac{1}{2}m(R+r)^2\dot{\theta}^2 + \frac{1}{2}I\dot{\phi}^2 \tag{10}$$

One more equation can be found by doing a linear momentum balance on the smaller cylinder in the $\bar{e}_r$.

LMB:

$$\Sigma F = m\bar{a}_G \tag{11}$$

$$-mg\hat{j} + N\hat{e}_r = m((R+r)\ddot{\theta}\hat{e}_\theta - (R+r)\dot{\theta}^2\hat{e}_r) \tag{12}$$

$$\{12\}\cdot\hat{e}_r \implies -mgcos(\theta) + N = m(-(R+r)\dot{\theta}^2) \tag{13}$$

This can be further reduced since $N = 0$ at the point where it leaves the other cylinder. The this equation can be solved for to to get $\dot{\theta}$ which can be substituted into the energy equation.

$$-mgcos(\theta) = m(-(R+r)\dot{\theta}^2) \tag{14}$$

$$\dot{\theta}^2 = \frac{gcos(\theta)}{R+r} \tag{15}$$

$$E^{TOT} = mg(R+r)cos(\theta) + \frac{1}{2}m(R+r)^2\left(\frac{gcos(\theta)}{R+r}\right) + \frac{1}{2}I\dot{\phi}^2 \tag{16}$$

$$E^{TOT} = mg(R+r)cos(\theta) + \frac{1}{2}mg(R+r)cos(\theta) + \frac{1}{2}I\dot{\phi}^2 \tag{17}$$

Now the only remaining unknown is ϕ but since the cylinder has no slip a relation can be found between ϕ and θ. We know that the velocity of the top cylinders center of mas can be found using either ϕ or θ, and they both must be equivalent until the cylinder leaves the bottom one.

$$r\phi = (R+r)\theta \tag{18}$$

$$r\dot{\phi} = (R+r)\dot{\theta} \tag{19}$$

Using this relation and equation 17 a relation for θ when they separate can be found.

$$E^{TOT} = mg(R+r)cos(\theta) + \frac{1}{2}mg(R+r)cos(\theta) + \frac{1}{2}I\left(\frac{(R+r)\dot{\theta}}{r}\right)^2 \tag{20}$$

$$E^{TOT} = mg(R+r)cos(\theta) + \frac{1}{2}mg(R+r)cos(\theta) + \frac{1}{2}I\left(\frac{(R+r)^2\frac{gcos(\theta)}{R+r}}{r^2}\right) \tag{21}$$

$$E^{TOT} = mg(R+r)cos(\theta) + \frac{1}{2}mg(R+r)cos(\theta) + \frac{1}{2}I\left(\frac{(R+r)gcos(\theta)}{r^2}\right) \tag{22}$$

The energy is conserved so the above equation can be set equal to the initial height and solved for in terms of θ.

$$mg(R+r) = mg(R+r)cos(\theta) + \frac{1}{2}mg(R+r)cos(\theta) + \frac{1}{2}I\left(\frac{(R+r)gcos(\theta)}{r^2}\right) \tag{23}$$

$$m = \frac{3}{2}mcos(\theta) + \frac{1}{2}I\left(\frac{cos(\theta)}{r^2}\right) \tag{24}$$

$$m = \left(\frac{3}{2}m + \frac{1}{2}\frac{I}{r^2}\right)cos(\theta) \tag{25}$$

$$\theta = cos^{-1}\left(\frac{m}{\frac{3}{2}m + \frac{1}{2}\frac{I}{r^2}}\right) \tag{26}$$